Shafollo Proshad Gupto

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Education

Bangladesh University of Engineering & Technology (BUET) *Bangladesh*

B.Sc. in Naval Architecture and Marine Engineering *Jan 2022 – Jun 2026*

CGPA: 3.77/4.00 • Ranked 4th among 55 students in the department

Relevant Courses: Fluid Mechanics • Marine Hydrodynamics • Computational Fluid Dynamics (CFD) • Numerical Computations Sessional • Resistance and Propulsion of Ships • Basic Thermal Engineering • Heat Transfer • Mechanics of Structure • Finite Element Method (FEM) for Ship Structure

Research Interests

- Computational Fluid Dynamics
- Heat Transfer
- Phase Change Materials

Research Experience

1. Independent collaborative research [View study] *2025–2026*

Research Title: ***Numerical Investigation of Melting Enhancement in a Phase Change Material-Filled Enclosure Using K-Shaped Fins: Effects of Fin Angle and Arm Length Ratio***

- Numerically investigated K-shaped aluminium fins for enhancing the melting of low-thermal-conductivity lauric acid PCM in a rectangular cavity using ANSYS Fluent 2024 R1.
- Evaluated five fin-arm-length configurations at fin angles of 10° to 80° and developed the Lower-zone Melting Lag Index (LMLI) to quantify thermal non-uniformity.
- Demonstrated up to 53.35% higher melting efficiency at 40°–50° with an extended lower fin arm and found an inverse relationship between melting efficiency and time-averaged LMLI.

2. Undergraduate Thesis [View thesis] *Apr 2025 – Jun 2026*

Thesis Title: ***Prediction of Calm-Water Resistance Using Smoothed Particle Hydrodynamics (SPH)***

Supervisor: Dr. Md. Mashiur Rahaman, Professor, BUET

- Investigated SPH for predicting the calm-water resistance of the KCS displacement hull.
- Implemented WCSPH using the GPU-accelerated DualSPHysics framework.
- Developed a numerical towing tank and identified 9 mm as the optimal particle spacing; resistance predictions at $Fr = 0.152$–0.282 showed errors below 6.5%.

Publications

Md Tanvir Anjum Zenith, Adri Das, **Shafollo Proshad Gupto**, Md. Atiqur Rahaman, Md. Khairul Islam, "Numerical Investigation of Melting Enhancement in a Phase Change Material-Filled Enclosure Using K-Shaped Fins: Effects of Fin Angle and Arm Length Ratio." *Case Studies in Thermal Engineering* **(Under Review)**.

Contribution: Conceptualization, Data Curation, Investigation, Software, Writing – Original Draft, Formal analysis.

Engineering Project

Undergraduate Project [View project] *Jan 2024 – Feb 2025*

Supervisor: Dr. Nayeb Md. Golam Zakaria

Design of an Inland General Cargo Vessel with 2500 Ton Cargo Capacity

Developed an inland general cargo ship with a 2,500-ton cargo carrying capacity for year-round operation on the Dhaka–Chittagong route. Performed detailed ship design, economic analysis, and assessment of environmental impacts.

Technical Skills

CAD & 3D Modelling	**Rhinoceros 3D, AutoCAD** Used both tools in the undergraduate ship-design project to model vessel geometry and draw ship components. Also prepared KCS hull geometry for numerical resistance simulations using Rhinoceros 3D.
Fluid Flow Analysis	**ANSYS Fluent, DualSPHysics, ParaView** Simulated PCM melting and natural convection in ANSYS Fluent; investigated KCS calm-water resistance using DualSPHysics; and post-processed and visualised simulation results in ParaView.
Programming & Numerical Computation	**Python, MATLAB, C++, Fortran** Used Python for research-data processing, filtering, calculations, and graph generation. Applied MATLAB to numerical ODE solutions and integration methods. Developed a C++ structured mesh generator as a course project, including node and element generation and element aspect-ratio and quality evaluation. Studied Fortran programming as part of the Computer Programming in Ship Design-I course.
Structural Analysis	**ABAQUS** Performed plane-stress, tangential-stress, compression, and buckling analyses of structural components, including truss systems, in Ship Structure Sessional.
Ship Design & Naval Architecture	**MAXSURF** Performed resistance, hydrostatic, intact and damage stability, and freeboard analyses for the undergraduate ship-design project and the WFSA ferry-design competition.

Professional Experience

Intern *Jan 2025 – Feb 2025*

Dockyard & Engineering Works Ltd, BANGLADESH NAVY, Narayanganj, Bangladesh

Completed shipyard practice at DEWBN as part of NAME 310, gaining practical exposure to shipbuilding processes, including welding, NDT inspections, material flow, pipe fabrication, and marine engine installation. Developed an understanding of ship production workflows and the practical constraints involved in translating engineering designs into physical marine structures.

Honors and Awards

- **University Dean's List Scholarship** (1st, 2nd and 4th year)
- **University Merit Scholarship** (1st, 2nd and 4th year)
- **3rd Prize** in "Worldwide Ferry Safety Association Design Competition 2026", Team Black Pearl. **[View Design Competition]**

Leadership and Service

- University Class Representative, Department of Naval Architecture and Marine Engineering, BUET *Dec 2022 – Jan 2025*
- Volunteer, MARTEC 2022 – International Conference on Marine Technology *Dec 2022*
- Team Member, Black Pearl, BUET, Worldwide Ferry Safety Association Design Competition *Sept 2025 – Mar 2026*

References

Dr. Nayeb Md. Golam Zakaria
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Relationship:
Ship design project supervisor.

Dr. Md. Mashiur Rahaman
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Relationship:
Undergraduate thesis supervisor.

Dr. Zobair Ibn Awal
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WFSA ferry design competition supervisor.